

## IV – Duality

For each model of Linear Programming with coefficients  $a_{ij}$ ,  $b_i$  and  $c_j$ , there is another model, named **dual**, formed by the same coefficients but arranged in a different way.

In the relationship with the **dual** problem, the initial problem is called **primal** problem.

The existence of one of the problems supports the existence of the other, being a reciprocal relationship of **duality**. In these conditions, both problems are known as a **pair of dual problems**.

Although the problems are different, they are supported by the same system of parameters ( $a_{ij}$ ,  $b_i$  and  $c_j$ ). The resolution of one of them allows the simultaneous resolution of the other, being the solution of each of them completely determined by the one of the other.

A pair of dual problems is, after all, a pairs of **representations of the same real problem**.

Consider the following **primal** problem in the typical form:

Determine

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix} \geq \mathbf{0}$$

in order to

$$\text{maximize } z_p = [c_1 \ c_2 \ \cdots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix}$$

subject to

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix} \leq \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ \vdots \\ b_m \end{bmatrix}$$

that is,

Determine

$$\mathbf{x} \geq \mathbf{0}$$

in order to

$$\text{maximize } z_p = \mathbf{c}'\mathbf{x}$$

subject to

$$\mathbf{Ax} \leq \mathbf{b}$$

Assigning to each **constraint i** of the **primal** problem the **variable  $u_i$**  of **dual**, and to each **variable  $x_j$**  of the **primal** problem the **constraint j** of the **dual**, the dual problem will be:

Determine

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \\ u_m \end{bmatrix} \geq \mathbf{0}$$

in order to

$$\text{minimize } z_d = [b_1 \ b_2 \ \dots \ b_m] \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \\ u_m \end{bmatrix}$$

subject to

$$\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ & \vdots & & \vdots \\ & \vdots & & \vdots \\ & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \geq \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

that is,

Determine

$$\mathbf{u} \geq \mathbf{0}$$

in order to

$$\text{minimize } z_d = \mathbf{b}'\mathbf{u}$$

subject to

$$\mathbf{A}'\mathbf{u} \geq \mathbf{c}$$

Note that:

- number of variables of the **dual** = number of constraints of the **primal**
- number of constraints of the **dual** = number of variables of the **primal**
- objective function of the **primal** in maximizing form  
→ objective function of the **dual** in minimizing form
- constraints of the **primal** of type  $\leq$   
→ variables of the **dual** non-negative
- variables of the **primal** non-negative  
→ constraints of the **dual** of type  $\geq$
- coefficients of the objective function of the **dual**  
= independent terms of the constraints of the **primal**
- independent terms of the constraints of the **dual**  
= coefficients of the objective function of the **primal**
- matrix of the coefficients of the variables in the constraints of the **dual**  
= transposed matrix of the coefficients of the variables in the constraints of the **primal**

**The dual of the dual is the primal**

## Example

Consider the example 1 of chapter II.

The **primal** problem is:

$$\text{maximize } z_p = 5 x_1 + 2 x_2$$

subject to

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2x_2 \leq 9$$

$$x_1 \geq 0, x_2 \geq 0$$

The corresponding **dual** problem will be:

$$\text{minimize } z_d = 3 u_1 + 4 u_2 + 9 u_3$$

subject to

$$u_1 + u_3 \geq 5$$

$$u_2 + 2 u_3 \geq 2$$

$$u_1 \geq 0, u_2 \geq 0, u_3 \geq 0$$

## Relationship Primal-Dual

| MAXIMIZATION<br>PROBLEM    | OBTAINING DUAL<br> |          | MINIMIZATION<br>PROBLEM    |
|----------------------------|------------------------------------------------------------------------------------------------------|----------|----------------------------|
| $i^{\text{th}}$ constraint | $\leq$                                                                                               | $\geq 0$ | $i^{\text{th}}$ variable   |
|                            | $\geq$                                                                                               | $\leq 0$ |                            |
|                            | $=$                                                                                                  | free     |                            |
| $j^{\text{th}}$ variable   | $\geq 0$                                                                                             | $\geq$   | $j^{\text{th}}$ constraint |
|                            | $\leq 0$                                                                                             | $\leq$   |                            |
|                            | free                                                                                                 | $=$      |                            |
| Matrix $A$                 | Matrix $A^T$                                                                                         |          |                            |
| Coefficients of the OF     | Independent terms                                                                                    |          |                            |
| Independent terms          | Coefficients of the OF                                                                               |          |                            |

## Fundamental properties of Duality

- If  $\mathbf{x}^*$  is the optimal solution of the **primal** problem and  $\mathbf{u}^*$  the optimal solution of the **dual** problem, thus  $z_p^* = \mathbf{c}'\mathbf{x}^*$  is equal to  $z_d^* = \mathbf{b}'\mathbf{u}^*$ . That is,  $z_p^* = z_d^*$ .
- A linear programming problem has a finite optimal solution if, and only if, there are feasible solutions for both **primal** and **dual** problems.
- If one of the problems has an unbounded optimal solution, then the other has no feasible solutions (it is impossible). If one of the problems has no feasible solutions (it is impossible) then the other also has no feasible solutions (it is also impossible) or it has an unbounded optimal solution.

## Example

Consider again the example of page IV-7.

The **primal** problem is:

$$\max z_p = 5 x_1 + 2 x_2$$

subject to

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2 x_2 \leq 9$$

$$x_1 \geq 0, x_2 \geq 0$$

The corresponding **dual** is:

$$\min z_d = 3 u_1 + 4 u_2 + 9 u_3$$

subject to

$$u_1 + u_3 \geq 5$$

$$u_2 + 2 u_3 \geq 2$$

$$u_1 \geq 0, u_2 \geq 0, u_3 \geq 0$$

Introducing the slack variables  $x_3$ ,  $x_4$  and  $x_5$  in the **primal** and the surplus variables  $u_4$  and  $u_5$  in the **dual**:

$$\max z_p = 5 x_1 + 2 x_2$$

subject to

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 4$$

$$x_1 + 2 x_2 + x_5 = 9$$

$$x_i \geq 0, i = 1, 2, \dots, 5$$

$$\min z_d = 3 u_1 + 4 u_2 + 9 u_3$$

subject to

$$u_1 + u_3 - u_4 = 5$$

$$u_2 + 2u_3 - u_5 = 2$$

$$u_i \geq 0, i = 1, 2, \dots, 5$$

Solving the **primal** problem by the Simplex method:

|                  |        | $c_i$ | 5          | 2          | 0          | 0          | 0          |     |
|------------------|--------|-------|------------|------------|------------|------------|------------|-----|
| $x_B$            | $c'_B$ | $x_i$ | $x_1$      | $x_2$      | $x_3$      | $x_4$      | $x_5$      | $b$ |
| $\leftarrow x_3$ | 0      |       | <u>1</u> * | 0          | 1          | 0          | 0          | 3   |
| $x_4$            | 0      |       | 0          | 1          | 0          | 1          | 0          | 4   |
| $x_5$            | 0      |       | 1          | 2          | 0          | 0          | 1          | 9   |
| $Z_j - c_j$      |        |       | -5         | -2         | 0          | 0          | 0          | 0   |
|                  |        |       | $\uparrow$ | $\uparrow$ | $\uparrow$ | $\uparrow$ | $\uparrow$ |     |
|                  |        |       | $u_4$      | $u_5$      | $u_1$      | $u_2$      | $u_3$      |     |

|                  |        | $c_i$ | 5          | 2          | 0          | 0          | 0          |     |
|------------------|--------|-------|------------|------------|------------|------------|------------|-----|
| $x_B$            | $c'_B$ | $x_i$ | $x_1$      | $x_2$      | $x_3$      | $x_4$      | $x_5$      | $b$ |
| $x_1$            | 5      |       | 1          | 0          | 1          | 0          | 0          | 3   |
| $x_4$            | 0      |       | 0          | 1          | 0          | 1          | 0          | 4   |
| $\leftarrow x_5$ | 0      |       | 0          | <u>2</u> * | -1         | 0          | 1          | 6   |
| $Z_j - c_j$      |        |       | 0          | -2         | 5          | 0          | 0          | 15  |
|                  |        |       | $\uparrow$ | $\uparrow$ | $\uparrow$ | $\uparrow$ | $\uparrow$ |     |
|                  |        |       | $u_4$      | $u_5$      | $u_1$      | $u_2$      | $u_3$      |     |

|             | $c_i$  | 5                     | 2                     | 0                     | 0                     | 0                     |                       |
|-------------|--------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| $x_B$       | $c'_B$ | $x_1$                 | $x_2$                 | $x_3$                 | $x_4$                 | $x_5$                 | $b$                   |
| $x_1$       | 5      | 1                     | 0                     | 1                     | 0                     | 0                     | 3                     |
| $x_4$       | 0      | 0                     | 0                     | $\frac{1}{2}$         | 1                     | $-\frac{1}{2}$        | 1                     |
| $x_2$       | 2      | 0                     | 1                     | $-\frac{1}{2}$        | 0                     | $\frac{1}{2}$         | 3                     |
| $Z_j - c_j$ |        | 0                     | 0                     | 4                     | 0                     | 1                     | 21 $\leftarrow Z_p^*$ |
|             |        | $\uparrow$<br>$u_4^*$ | $\uparrow$<br>$u_5^*$ | $\uparrow$<br>$u_1^*$ | $\uparrow$<br>$u_2^*$ | $\uparrow$<br>$u_3^*$ | $\uparrow$<br>$Z_d^*$ |

Optimal solution of the primal problem:  $x^* = (3, 3, 0, 1, 0) \longrightarrow z_p^* = 21$

Through the row “ $z_j - c_j$ ” of the optimal table of the **primal** problem, we are able to determine the optimal solution of the **dual** problem:  $u^* = (4, 0, 1, 0, 0) \longrightarrow z_d^* = z_p^* = 21$

In effect, in each iteration:

- ☛ The value of the original variable  $u_i$  of the **dual** is equal to the coefficient of the variable slack/surplus  $x_{n+i}$  of the **primal** in row “ $z_j - c_j$ ”;
- ☛ The value of the variable slack/surplus  $u_{m+j}$  of the **dual** is equal to the coefficient of the original variable  $x_j$  of the **primal** in row “ $z_j - c_j$ ”.

## Complementary basic solutions property

Each basic solution of the **primal** problem has a complementary basic solution in the dual problem.

## Complementary slackness property

The variables in complementary basic solutions of **primal** and **dual** problems satisfy the following relationships:

| <i>primal variable</i> |   | <i>associated dual variable</i> |
|------------------------|---|---------------------------------|
| basic                  | → | non-basic                       |
| non-basic              | → | basic                           |

Although the complementary slackness property is verified for each pair of complementary basic solutions of **primal-dual** problems, only when the optimal is reached the solutions are both feasible.

Consider again the example presented in page IV-12.

- In the first simplex table the solutions of the primal and dual problems are:  $\mathbf{x} = (0, 0, 3, 4, 9)$  and  $\mathbf{u} = (0, 0, 0, -5, -2)$ . That is, the solution of the **primal** is feasible but the one of the **dual** is not.
- In the second Simplex table, the solutions are:  $\mathbf{x} = (3, 0, 0, 4, 6)$  and  $\mathbf{u} = (5, 0, 0, 0, -2)$ . The situation in terms of feasibility is the same as before.
- Finally, in the third Simplex table that is optimal, the solutions are both feasible:  $\mathbf{x}^* = (3, 3, 0, 1, 0)$  and  $\mathbf{u}^* = (4, 0, 1, 0, 0)$ .

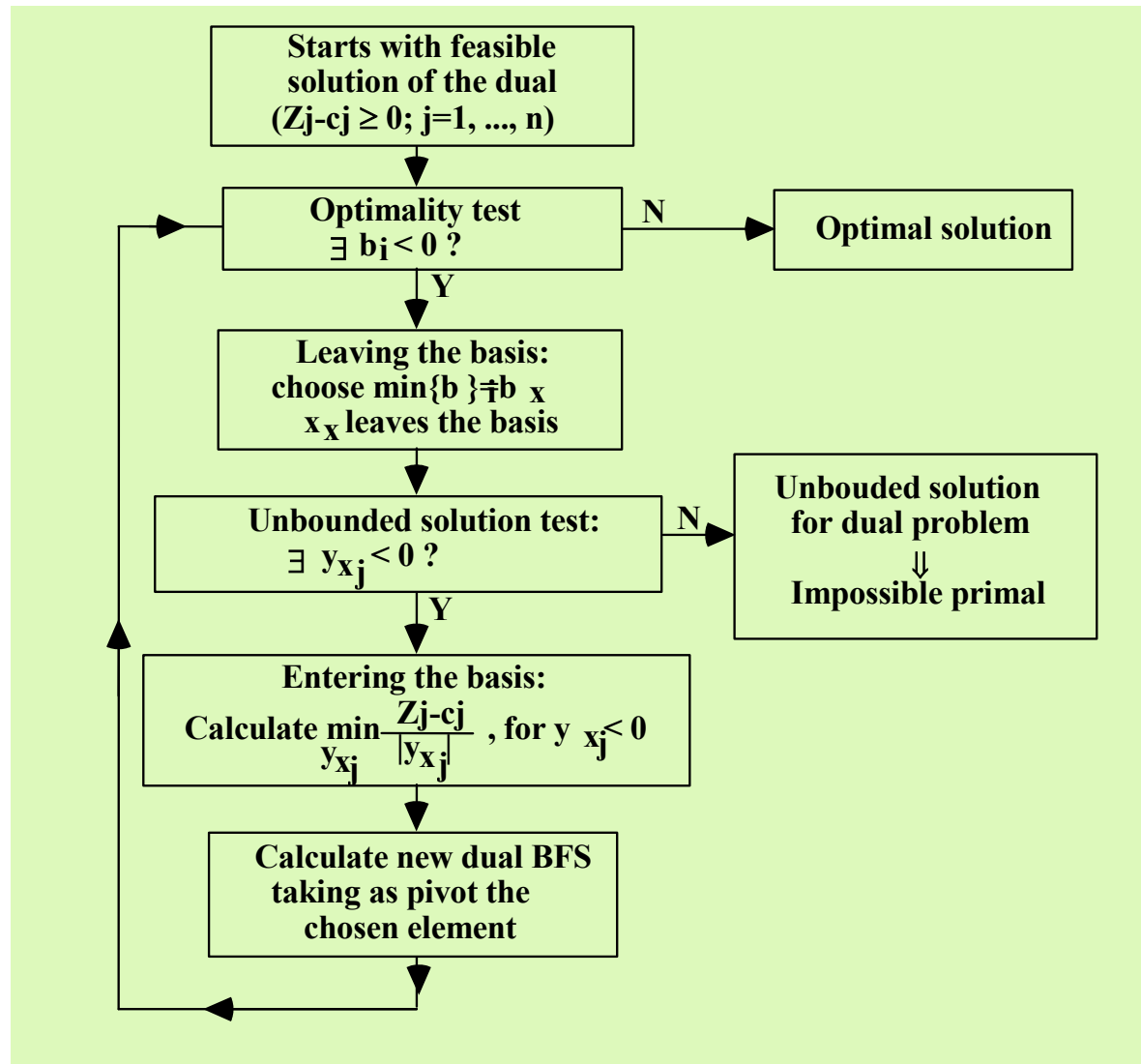
## Dual Simplex method

The **Simplex method** moves from feasible solution to feasible solution of the **primal** (and, correspondingly, from not feasible solution to not feasible solution of the **dual**) until find a pair of feasible solutions to the **primal** and **dual** problems, which optimize the objective functions.

The **dual simplex method** starts from a feasible solution of the **dual** (and, correspondingly, not feasible of the **primal**) and proceeds iteratively until find a pair of feasible solutions to the **primal** and **dual** problems, or until conclude that the **dual** problem has no bounded solution (and the **primal** problem has no solution, it is impossible).

It means that, while the **Simplex method** maintains the **primal** solution feasibility, the **dual Simplex method** maintains the **dual** solution feasibility. For this reason, it is said that the latter method is the "dual image" of the first.

## Flowchart of dual Simplex method



## Example

Consider the following **primal** and associated **dual** problems:

$$\min z_p = 3 x_1 + 4 x_2 + 9 x_3$$

subject to

$$x_1 + x_3 \geq 5$$

$$x_2 + 2x_3 \geq 2$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

$$\max z_d = 5 u_1 + 2 u_2$$

subject to

$$u_1 \leq 3$$

$$u_2 \leq 4$$

$$u_1 + 2 u_2 \leq 9$$

$$u_1 \geq 0, u_2 \geq 0$$

Making the necessary changes to the **primal** in order to create negative values in the independent terms:

$$\max z_p' = -z_p = -3 x_1 - 4 x_2 - 9 x_3$$

subject to

$$-x_1 - x_3 + x_4 = -5$$

$$-x_2 - 2x_3 + x_5 = -2$$

$$x_i \geq 0; i = 1, \dots, 5.$$

Solving the **primal** problem by the dual Simplex method:

{follow this resolution with pages III-12 and III-13}

|                  | $c_i$  | -3          | -4         | -9         | 0          | 0          |                     |
|------------------|--------|-------------|------------|------------|------------|------------|---------------------|
| $x_B$            | $c'_B$ | $x_1$       | $x_2$      | $x_3$      | $x_4$      | $x_5$      | $b$                 |
| $\leftarrow x_4$ | 0      | <u>-1</u> * | 0          | -1         | 1          | 0          | -5 $\leftarrow x_4$ |
| $x_5$            | 0      | 0           | -1         | -2         | 0          | 1          | -2 $\leftarrow x_5$ |
| $Z_j - c_j$      |        | 3           | 4          | 9          | 0          | 0          | 0 $\leftarrow -z_p$ |
|                  |        | $\uparrow$  | $\uparrow$ | $\uparrow$ | $\uparrow$ | $\uparrow$ |                     |
|                  |        | $u_3$       | $u_4$      | $u_5$      | $u_1$      | $u_2$      |                     |

NFBSP  $\rightarrow x = (0, 0, 0, -5, -2)$

FBSD  $\rightarrow u = (0, 0, 3, 4, 9)$

|                  | $c_i$  | -3         | -4         | -9          | 0          | 0          |                       |
|------------------|--------|------------|------------|-------------|------------|------------|-----------------------|
| $x_B$            | $c'_B$ | $x_1$      | $x_2$      | $x_3$       | $x_4$      | $x_5$      | $b$                   |
| $x_1$            | -3     | 1          | 0          | 1           | -1         | 0          | 5 $\leftarrow x_1$    |
| $\leftarrow x_5$ | 0      | 0          | -1         | <u>-2</u> * | 0          | 1          | -2 $\leftarrow x_5$   |
| $Z_j - c_j$      |        | 0          | 4          | 6           | 3          | 0          | -15 $\leftarrow -z_p$ |
|                  |        | $\uparrow$ | $\uparrow$ | $\uparrow$  | $\uparrow$ | $\uparrow$ |                       |
|                  |        | $u_3$      | $u_4$      | $u_5$       | $u_1$      | $u_2$      |                       |

NFBSP  $\rightarrow x = (5, 0, 0, 0, -2)$

FBSD  $\rightarrow u = (3, 0, 0, 4, 6)$

|                                      | <b>c<sub>i</sub></b>                | <b>-3</b>                        | <b>-4</b>                        | <b>-9</b>                        | <b>0</b>                         | <b>0</b>                         |                                   |
|--------------------------------------|-------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| <b>x<sub>B</sub></b>                 | <b>c'<sub>B</sub> x<sub>i</sub></b> | <b>x<sub>1</sub></b>             | <b>x<sub>2</sub></b>             | <b>x<sub>3</sub></b>             | <b>x<sub>4</sub></b>             | <b>x<sub>5</sub></b>             | <b>b</b>                          |
| <b>x<sub>1</sub></b>                 | <b>-3</b>                           | <b>1</b>                         | <b>-1/2</b>                      | <b>0</b>                         | <b>-1</b>                        | <b>1/2</b>                       | <b>4</b> $\leftarrow x_1^*$       |
| <b>x<sub>3</sub></b>                 | <b>-9</b>                           | <b>0</b>                         | <b>1/2</b>                       | <b>1</b>                         | <b>0</b>                         | <b>-1/2</b>                      | <b>1</b> $\leftarrow x_3^*$       |
| <b>Z<sub>j</sub> - c<sub>j</sub></b> |                                     | <b>0</b>                         | <b>1</b>                         | <b>0</b>                         | <b>3</b>                         | <b>3</b>                         | <b>-21</b> $\leftarrow -z_p^*$    |
|                                      |                                     | <b>↑</b>                         | <b>↑</b>                         | <b>↑</b>                         | <b>↑</b>                         | <b>↑</b>                         | <b>↑</b>                          |
|                                      |                                     | <b>u<sub>3</sub><sup>*</sup></b> | <b>u<sub>4</sub><sup>*</sup></b> | <b>u<sub>5</sub><sup>*</sup></b> | <b>u<sub>1</sub><sup>*</sup></b> | <b>u<sub>2</sub><sup>*</sup></b> | <b>-z<sub>d</sub><sup>*</sup></b> |

Optimal table since there aren't negative values in the b column.

Optimal solutions of primal and dual problems:

$$\text{FBSP} \rightarrow \mathbf{x}^* = (4, 0, 1, 0, 0)$$

$$\text{FBSD} \rightarrow \mathbf{u}^* = (3, 3, 0, 1, 0)$$